

1. [10 points] Determine the exact value if these series converge or state why it diverges.

$$a) \sum_{n=2}^{\infty} \frac{1-2^n}{3^n}$$

$-\frac{7}{6}$ and converges

$$\sum_{n=2}^{\infty} \left(\frac{1}{3}\right)^n - \sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n \Rightarrow \sum_{n=2}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^n - \sum_{n=2}^{\infty} \frac{4}{9} \left(\frac{2}{3}\right)^n$$

$$\sum_{n=2}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^n$$

$$\sum_{n=2}^{\infty} \frac{4}{9} \left(\frac{2}{3}\right)^n$$

$$\frac{1}{6} - \frac{4}{3} = \boxed{-\frac{7}{6}}$$

both converge

(+6) $r = 0.\bar{3} \Rightarrow -1 < r = 0.\bar{3} < 1$
converges by the
geometric test

$$\frac{a}{1-r} = \frac{1/9}{1-1/3} = \frac{1/9}{2/3} = \boxed{\frac{1}{6}}$$

$r = 0.\bar{6} \Rightarrow -1 < r = 0.\bar{6} < 1$
converges by
geometric test

$$\frac{a}{1-r} = \frac{4/9}{1-2/3} = \frac{4/9}{1/3} = \boxed{\frac{4}{3}}$$

✓
V. good work!

b. $\sum_{n=1}^{\infty} (\sqrt{n+2} - \sqrt{n})$

0 and converges

$$S_1 = a_1 = (\sqrt{1+2} - \sqrt{1})$$

$$S_2 = a_1 + a_2 = (\sqrt{1+2} - \sqrt{1}) + (\sqrt{2+2} - \sqrt{2})$$

$$S_3 = a_1 + a_2 + a_3 = (\sqrt{1+2} - \sqrt{1}) + (\sqrt{2+2} - \sqrt{2}) + (\sqrt{3+2} - \sqrt{3})$$

$$S_4 = a_1 + a_2 + a_3 + a_4 = (\sqrt{1+2} - \sqrt{1}) + (\sqrt{2+2} - \sqrt{2}) + (\sqrt{3+2} - \sqrt{3}) + (\sqrt{4+2} - \sqrt{4})$$

$$S_n = (\cancel{\sqrt{1+2}} - \sqrt{1}) + (\cancel{\sqrt{2+2}} - \sqrt{2})$$

$$(\sqrt{n+1} - \sqrt{1}) + (\sqrt{n+2} - \sqrt{2})$$

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{1}) + (\sqrt{n+2} - \sqrt{2}) = \infty + \infty$$

LH#1

$$\lim_{n \rightarrow \infty} \frac{-\frac{1}{2\sqrt{n+1}}}{\frac{1}{2\sqrt{n+2}}} = 0$$

Good work!

2. [11 points] Given $\sum_{n=1}^{\infty} \frac{(-1)^n}{3+\sqrt{n}}$

a. Use appropriate test to determine if the series $\sum_{n=1}^{\infty} |a_n|$ converge or diverge. diverges

$$|a_n| = \left| \frac{(-1)^n}{3+\sqrt{n}} \right| \Rightarrow \frac{1}{3+\sqrt{n}}$$

$$a_n = \frac{1}{3+\sqrt{n}}$$

• Comparison test was applied first then limit comparison

V. good work!

$$b_n = \frac{1}{\sqrt{n}} =$$

$$p = 0.5$$

$$p = 0.5 < 1$$

b_n diverges by p-series

$|a_n|$ diverges

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} \Rightarrow \frac{1}{3+\sqrt{n}} \cdot \sqrt{n} = \frac{\sqrt{n}}{3+\sqrt{n}} = \frac{\infty}{\infty} \text{ = apply L'Hospital rule}$$

$$L \neq 1 \quad \lim_{n \rightarrow \infty} \frac{\frac{1}{2\sqrt{n}}}{\frac{1}{2\sqrt{n}}} = 1$$

$$0 < L < \infty$$

a_n diverges by limit comparison

b. Use appropriate test to determine if the series $\sum_{n=1}^{\infty} a_n$ converge or diverge. converges

$$a_n = \frac{(-1)^n}{3+\sqrt{n}}$$

$$b_n = \frac{1}{3+\sqrt{n}} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{3+\sqrt{n}} = 0 \quad \checkmark$$

Not true

$$b_n = (3+\sqrt{n})^{-1}$$

(+3)

$$\Rightarrow \frac{d}{dn} ((3^{-1} + n^{-1/2})) \Rightarrow -\frac{1}{2n^{3/2}} \text{ (decreasing)}$$

a_n converges by the alternating series test

c. Explain why $\sum_{n=1}^{\infty} a_n$ is convergent, conditionally convergent, or divergent

$\sum_{n=1}^{\infty} a_n$ conditionally converges because $|a_n|$ diverges

(+1)

while a_n converges

3. [12 points] Determine if the following series converge or diverge using appropriate tests.

a. $\sum_{n=1}^{\infty} \frac{n!}{5^n}$

diverges

$a_n = \frac{n!}{5^n}$

V. good work!

- $0 \leq r < 1$ abs converges
- $r = 1$ test fails
- $r > 1$ diverges

$a_{n+1} = \frac{n! (n+1)}{5^{n+1}} \Rightarrow \frac{n! (n+1)}{5^n \cdot 5}$

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \Rightarrow \left| \frac{n! (n+1)}{5^n \cdot 5} \cdot \frac{5^n}{n!} \right| \Rightarrow \lim_{n \rightarrow \infty} \frac{(n+1)}{5} = \infty$

(+5) diverges by the ratio test

b. $\sum_{n=1}^{\infty} \left(\frac{5e^n + 1}{e^n + n} \right)^n$

diverges

$\sqrt[n]{\left| \left(\frac{5e^n + 1}{e^n + n} \right) \right|}$

$\Rightarrow \lim_{n \rightarrow \infty} \frac{5e^n + 1}{e^n + n} = \frac{\infty}{\infty}$

• apply L'Hospital's rule

LH #1 $\lim_{n \rightarrow \infty} \frac{5e^n}{e^n + 1} = \frac{\infty}{\infty}$

• apply L'Hospital's rule

V. good work!

LH #2 $\lim_{n \rightarrow \infty} \frac{5e^n}{e^n} = 5$

- $0 \leq r < 1$ abs converges
- $r = 1$ test fails
- $r > 1$ diverges

(+5)

diverges by the root test

c. $\sum_{n=1}^{\infty} (2n+1)$

diverges

$a_n = \frac{1}{(2n+1)^{-1}}$

$b_n = 2n = \frac{2}{n^{-1}}$

$b_n \leq a_n$

$\frac{1}{2n^{-1}} \leq \frac{1}{(2n+1)^{-1}} \Rightarrow 2n \leq 2n+1$

$0 \leq 1$
true

$b_n = \frac{1}{2n^{-1}} = \frac{1}{2n^{-1}}$

Wrong b_n . b_n diverges by

P-series test

a_n diverges by the comparison test

(+1)

$b_n = \frac{2}{n^{-1}}$

4. [10 points] Use two different test to determine if the series converge or diverge $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges

test #1 P-series

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

$$p=3$$

$$p=3 > 1$$

converges by the P-series test ✓

test #2 integral test

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

✓ Positive

$x \geq 1$ all Positive

$$f(x) = \frac{1}{x^3} > 0$$

$$\rightarrow f(x) = \frac{1}{x^3}$$

✓ Continuous

$$n^3 \neq 0$$

$$\sqrt[3]{n^3} = \sqrt[3]{0}$$

$$n \neq 0$$

$$(-\infty, 0) \cup (0, \infty)$$

continuous for all $x \geq 1$ ✓ values

✓ decreasing

$$\frac{d}{dn} \left(\frac{1}{n^3} \right) \Rightarrow -3 \frac{1}{n^4}$$

$$\Rightarrow \sqrt{-\frac{3}{n^4}}$$

$$\int_1^{\infty} \frac{1}{n^3} dn \Rightarrow$$

$$\lim_{m \rightarrow \infty}$$

$$\int_1^m$$

$$\frac{1}{n^3} dn \approx$$

$$\int_1^m \frac{1}{n^3} = \left[-\frac{1}{2n^2} \right]_1^m$$

$$\lim_{m \rightarrow \infty}$$

$$\left[-\frac{1}{2n^2} \right]_1^m$$

$$\Rightarrow$$

$$\lim_{m \rightarrow \infty}$$

$$\left[-\frac{1}{2m^2} - \left(-\frac{1}{2(1)^2} \right) \right]$$

$$= \frac{1}{2}$$

converges by the integral test

5. [5 points] Evaluate the integral using appropriate techniques: $\int_0^4 \frac{1}{\sqrt{4-x}} dx$ -4

$$\int_0^4 \frac{1}{\sqrt{4-x}} dx = \lim_{m \rightarrow 4^-} \int_0^m \frac{1}{\sqrt{4-x}} dx \approx \int_0^m \frac{1}{(u)^{1/2}} \frac{dx}{du} du$$

$x = 4 - u$
 $du = -dx$
 $= -du - dx$

$$\left[-2 \cdot u^{1/2} \right]_0^m \Rightarrow \lim_{m \rightarrow 4^-} \left[-2 \cdot (4-x)^{1/2} \right]_0^m$$

$$\lim_{m \rightarrow 4^-} \left[(2 \cdot (4-m)^{1/2}) - (2 \cdot (4-0)^{1/2}) \right]$$

$$\Rightarrow -2(4)^{1/2} = \boxed{-4}$$

(+4)

6. [5 points] Evaluate the limit using appropriate techniques: $\lim_{x \rightarrow 0^+} (\sin x)^x$ $e^0 \approx 1$

$$l = \lim_{x \rightarrow 0^+} (\sin x)^x \Rightarrow \ln(l) = \lim_{x \rightarrow 0^+} x \cdot \ln(\sin x)$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(\sin x)}{x^{-1}} = \frac{0}{0} \quad \bullet \text{ apply L'Hospital's rule}$$

$$\text{L'H} \quad \lim_{x \rightarrow 0^+} - \frac{\cos x}{\sin x} \cdot \frac{1}{x^{-2}} \approx \lim_{x \rightarrow 0^+} - \frac{\cos x}{\sin x} \cdot x^2 = \frac{0}{0}$$

✓ L'H

$$\ln(l) = 0$$

$$e^{\ln(l)} = e^0$$

$$l = e^0 \approx \boxed{l = 1}$$

(x3)